



Final Report

Title: Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)

Calvert Study No.: 1275RL35.001

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

Date: 23 Dec 2014

LABORATORY
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447 USA
Phone: 570.586.2411
Fax: 570.586.3450
www.calvertlabs.com

CORPORATE
1225 Crescent Green Dr.
Suite 115
Cary, NC 27518
Phone: 919.854.4453
Fax: 919.854.2860
www.calvertholdings.com

I. Table of Contents

I.	Table of Contents.....	2
II.	List of Abbreviations.....	4
III.	Regulatory Compliance and Signatures.....	5
A.	Regulatory Compliance	5
1.	Calvert Laboratories-Conducted Study Elements.....	5
2.	Guidelines for Study Design	5
B.	Study Director Signature.....	6
C.	Signatures of Other Responsible Personnel.....	6
IV.	Quality Assurance Unit Statement	7
A.	Protocol Inspection	7
B.	Protocol Amendment	7
C.	Procedures	7
D.	Data/Draft Report.....	7
V.	Summary.....	8
A.	Title.....	8
B.	Objective.....	8
C.	Methods.....	8
D.	Results.....	9
E.	Conclusion.....	9
VI.	General Information	10
A.	Key Study Dates.....	10
B.	Responsible Personnel Testing Facility	10
C.	Contributing Scientists	10
D.	Objective.....	11
E.	Experimental Design Overview.....	11
1.	General Description.....	11
VII.	Materials and Methods.....	11
A.	Test Article, Vehicle and Preparation.....	11
1.	Test Article.....	11
2.	Vehicle.....	12
3.	Test Article Preparation	12
4.	Dose Formulation Samples	14
5.	Dose Formulation Analysis	14
6.	Sample Batch Acceptance Criteria	16
7.	Reserve Archive Samples	17
8.	Accountability and Disposition.....	17
B.	Test System (Animals and Animal Care).....	17
1.	Description.....	17
2.	Rationale for Choice of Species and Number of Animals	17
3.	Husbandry	17
4.	Pre-study Health Screen and Selection Criteria	18
5.	Assignment to Study Groups	19
6.	Humane Care of Animals.....	19
C.	Test Article Administration	19
1.	Group Assignments and Dose Levels.....	19

2. Dosing	19
3. Justification for Route and Dose Levels.....	20
D. Method of Study Performance	20
E. Terminal Procedures	21
1. Termination.....	21
VIII. Records and Reports	21
A. Data Collection and Statistical Analysis.....	21
B. Storage of Records.....	21
IX. Results	22
A. Respiratory Parameters.....	22
1. Effects of Vehicle.....	22
2. Effects of LT3114.....	22
X. Conclusion	23
XI. References.....	23
XII. Table	24
A. Table 1- Evaluation of Respiratory Function of LT3114 in Male Rats (n=6) .	25
XIII. Appendices	26
A. Appendix I - Individual Data	
B. Appendix II - Test Article Information	
C. Appendix III - Test Article Formulation Analyses	
D. Appendix IV – Food and Water Analysis	
E. Appendix V – Protocol, Protocol Amendments and Deviations	

II. List of Abbreviations

CFR	Code of Federal Regulations
C of A	Certificate of Analysis
CO ₂	Carbon Dioxide
CS	Calibration Standard
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practice
HPLC	High-performance liquid chromatography
IACUC	Institutional Animal Care and Use Committee
ICH	International Conference on Harmonisation
ILAR	Institute for Laboratory Animal Resources
OECD	Organisation of Economic Co-operation and Development
QC	Quality Control
SOP	Standard Operating Procedure
USDA	United States Department of Agriculture

III. Regulatory Compliance and Signatures

Calvert Study No.: 1275RL35.001

Title: Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)

A. Regulatory Compliance

1. Calvert Laboratories-Conducted Study Elements

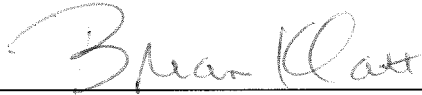
All parts of this study were performed according to SOP's and the protocol and conformed to the U.S. FDA GLP regulations in effect at the time of study conduct (Good Laboratory Practice Regulations, Title 21 Code of Federal Regulations Part 58) and in accordance with OECD Principles of Good Laboratory Practice as revised on 26th November, 1997 [C(97)186/Final].

2. Guidelines for Study Design

FDA Guidance for Industry "S7A Safety Pharmacology Studies for Human Pharmaceuticals (ICH)" issued July 2001

B. Study Director Signature

I, the undersigned, hereby declare that this report is a true and accurate record of the results obtained. No circumstances occurred during the study that were considered to have significantly affected the overall quality or integrity of the data obtained. All protocol deviations were documented in the study records and are listed in Appendix V of this report.



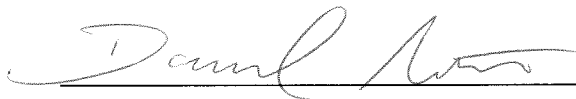
Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.



Date

C. Signatures of Other Responsible Personnel

Scientific Oversight and Report Review



Scientific Management
Calvert Laboratories, Inc.



Date

IV. Quality Assurance Unit Statement

Calvert Study No.: 1275RL35.001

Title: Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)

The following inspections were made in accordance with appropriate Calvert Standard Operating Procedures.

A. Protocol Inspection

Date of Inspection	Report to Study Director	Report to Management
6 Aug 2014	7 Aug 2014	7 Aug 2014

B. Protocol Amendment

Amendment	Date of Inspection	Report to Study Director	Report to Management
1	21 Aug 2014	22 Aug 2014	22 Aug 2014
2	3 Nov 2014	3 Nov 2014	3 Nov 2014

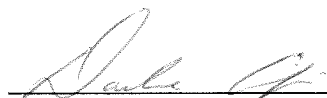
C. Procedures

Inspected Phase	Date of Inspection	Report to Study Director	Report to Management
Test Article Prep	13 Aug 2014	13 Aug 2014	13 Aug 2014

D. Data/Draft Report

Inspected Document	Date of Inspection	Report to Study Director	Report to Management
Data/Draft Report #1	16-17 Sep 2014	17 Sep 2014	17 Sep 2014
Draft Report #2	28 Oct 2014	28 Oct 2014	28 Oct 2014
Final Report	23 Dec 2014	23 Dec 2014	23 Dec 2014

This report has been reviewed by the Quality Assurance Unit, employing methods detailed in appropriate Calvert Standard Operating Procedures. The results constitute an accurate representation of recorded data. Any data supplied by the Sponsor/subcontractor were not audited by the Quality Assurance Unit at Calvert. Refer to individual Principal Investigator reports included in the appendices of this report for Quality Assurance Statements from test sites.



Darlene Gilpin, RQAP-GLP
Director, Quality Assurance
Calvert Laboratories, Inc.

23 Dec 2014
Date

V. Summary

A. Title

Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)

B. Objective

The purpose of this study was to determine the potential effects of LT3114 administered intravenously on respiratory rate, tidal volume and minute volume in the conscious rat.

C. Methods

Twenty four male rats were trained on two days in the head-out plethysmographic chamber for approximately 15 minutes each, before the day of the experiment. On the day of dosing, each animal was placed in its plethysmographic chamber and baseline values were obtained for 5 minutes following an approximately 5-minute stabilization period. The rats were then removed from the chamber and dosed by slow intravenous injection (~1 minute) as per the following schedule:

Group	Intravenous Treatment	Dose (mg/kg)	Volume (mL/kg)	Concentration (mg/mL)	No. of Animals
					Male
1	Vehicle control (Diluent)	0	5	0	6
2	LT3114	10	5	2.0	6
3	LT3114	30	5	6	6
4	LT3114	50	5	10	6

Following dosing, each animal was returned to its designated plethysmograph chamber and the respiratory parameters were recorded at 15 minutes (± 3 min), 1, 2, and 4 hours (± 5 minutes) following dose administration of test article/vehicle. Animals were allowed to stabilize in the plethysmograph for at least 5 minutes before each reading was taken. The following parameters were acquired, recorded and analyzed using DSI Dataquest Open ART (v. 2.3) & PONEMAH Physiology Platform (v. 4.2.0).

- Tidal Volume
- Minute Volume
- Respiratory rate

Values for the test article treated group were compared back to the vehicle controls and baseline using a two-way repeated measures ANOVA followed by a Bonferroni

Multiple Comparison Test (SigmaStat, v. 2.03). Differences with p values ≤ 0.05 were considered statistically significant.

D. Results

Effects of Vehicle:

The intravenous administration of the vehicle, LT3114 Diluent, at 5 mL/kg did not produce any statistically significant changes in respiratory rate or tidal volume compared to baseline values in male rats.

A statistically significant ($p \leq 0.05$) decrease (14%) in minute volume was observed at 1 hour following dosing of the vehicle when compared to the predose (baseline) values. Sporadic changes like this are occasionally observed in this model and not considered biologically relevant.

Effects of LT3114:

The intravenous administration of LT3114 at 10 mg/kg produced a statistically significant ($p \leq 0.05$) decrease (19%) in respiratory rate at 4 hours following dosing when compared to pre-dose values. The decrease in respiratory rate was not statistically significant when compared to the vehicle control group. The decrease in respiratory rate observed in the 10 mg/kg dose group at 4 hours post-dose was not dose-dependent at the higher doses and therefore not considered test article related. Minute volume and tidal volume were not affected by the intravenous administration of LT3114 at 10 mg/kg.

Statistically significant ($p \leq 0.05$) decreases (8%) in minute volume were observed at 4 hours following the intravenous administration of LT3114 at 30 and 50 mg/kg compared to the vehicle control group values. The changes were not statistically significant compared to the baseline values. Also the predose minute volume values were somewhat higher in the vehicle group than in the test article groups especially in Groups 3 and 4. Since the magnitude of the changes in minute volume was similar to the magnitude of changes occasionally observed in the vehicle control group, these changes were not considered biologically relevant. Respiratory rate and tidal volume were not affected by the 30 or 50 mg/kg doses of LT3114.

E. Conclusion

The intravenous administration of LT3114 at 10, 30 and 50 mg/kg did not induce any biologically relevant effects on respiratory rate, tidal volume or minute volume in conscious male rats.

VI. General Information

A. Key Study Dates

Study Initiation Date:	1 Aug 2014
Animal Receipt Date:	6 Aug 2014
Experimental Start Date:	13 Aug 2014
Experimental Completion Date:	25 Aug 2014

B. Responsible Personnel Testing Facility

Study Director:	Brian Klatt, B.S.
Primary Technicians:	Brian Klatt, B.S. Jennifer Shenise, A.S., ALAT Maura Boyarsky
Pharmacy Technicians:	Katelynn Henrich, B.S.

C. Contributing Scientists

Principal Investigator	
Dosing Formulation Analyses:	Victor J. Johnson, Ph.D. Grants & Contracts PI/Study Director Burleson Research Technologies 120 First Flight Lane Morrisville, NC 27560 Phone: (919) 719-2500 Fax: (866) 971-2128 Email: vjohnson@brt-labs.com
Study Monitor:	Marlene W. Modi, Ph.D. MWM Consulting Group, Inc. for Lpath Inc. 14 Joseph Court Monmouth Junction, NJ 08852 Phone: (732) 274-9562 Mobile: (732) 822-6101 E-mail: mwmconsultinggroup@gmail.com
Sponsor Representative:	Gary Woodnutt, PhD

Senior Vice President, Research
Lpath Inc
4025 Sorrento Valley Blvd
San Diego, CA 92121
Phone: (858) 678-0800 ext 112
Email: gwoodnutt@lpath.com

D. Objective

The purpose of this study was to determine the potential effects of LT3114 administered intravenously on respiratory rate, tidal volume and minute volume in the conscious rat.

E. Experimental Design Overview

1. General Description

Three groups of six rats were intravenously administered LT3114 at 10, 30 and 50 mg/kg at a dose volume of 5 mL/kg. An additional group of six rats was administered the vehicle (LT3114 Diluent) by intravenous injection at 5 mL/kg. Following dosing, each animal was returned to its designated plethysmograph chamber and the respiratory parameters were recorded 15 minutes (± 3 min), 1, 2, and 4 hours (± 5 minutes) following the administration of test article/vehicle. Animals were allowed to stabilize in the plethysmograph for at least 5 minutes before each reading was taken. Tidal volume, minute volume and respiratory rate were acquired and recorded using DSI Dataquest Open ART (v. 2.3) & PONEMAH Physiology Platform (v. 4.20).

VII. Materials and Methods

A. Test Article, Vehicle and Preparation

1. Test Article

Identification:	LT3114 (Bulk Drug Substance, Antibody LT-3114)
Lot No.:	103948
Physical Description:	Clear, colorless liquid
Storage Conditions:	Refrigerated (4-7°C) and protected from light.

Characterization and

Stability: A Certificate of Analysis was provided and is included in Appendix II. Three month stability is provided in Appendix II.

2. Vehicle

Identification: LT3114 Diluent [24.3 mM Citrate, 51.4 mM Sodium Phosphate, 300 mM D-trehalose pH 5.0 (DF Buffer)]

Lot/Batch No.: 104011-001

Physical Description: Clear colorless liquid

Storage Conditions: 4-7°C

Preparation: The diluent and Polysorbate 80 were removed from the refrigerator and retained at room temperature for at least 30 minutes prior to preparation. All procedures were carried out in a cleaned laminar flow hood using aseptic techniques whenever possible. For each bag of diluent, 2 mL of Polysorbate 80 was measured using a sterile 3 mL syringe and sterile 18 gauge needle and injected directly into the bag of vehicle. The bag was gently inverted until the solution appeared homogeneous. Once the bags were prepared, they were removed from the cleaned laminar flow hood and returned to their appropriate storage condition.

3. Test Article Preparation

The Sponsor provided the test article already formulated at 32.1 mg/ml in sterile containers. Diluent was also provided in separate sterile containers. Diluent and test article were allowed to come to room temperature before the formulation was prepared. The formulation was not shaken. Group 1 control animals received the diluent. When necessary depending on the dose level, doses were prepared by dilution using the provided 32.1 mg/ml and diluent solutions under aseptic conditions. Before dose preparation, 2

mL of the 10% Polysorbate 80 solution provided, was added aseptically to each liter of formulation buffer, also supplied, and inverted gently to ensure distribution throughout the volume of buffer. This diluent was then used for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations.

The appropriate volume of LT3114 stock solution was diluted in the appropriate volume of vehicle to achieve final nominal concentration dosing concentrations on a mg/ml basis as follows:

Date	Dose concentration	Amount of LT3114 32.1 mg/mL Stock	Volume of Vehicle	Description (pH)
13 Aug 2014	2 mg/mL	0.62 mL	9.38 mL	Clear, colorless liquid (6)
	6 mg/mL	1.87 mL	8.13 mL	Clear, colorless liquid (5)
	10 mg/mL	3.12 mL	6.88 mL	Clear, colorless liquid (5)
14 Aug 2014	2 mg/mL	0.31 mL	4.69 mL	Clear, colorless liquid
	6 mg/mL	0.93 mL	4.07 mL	Clear, colorless liquid
	10 mg/mL	1.56 mL	3.44 mL	Clear, colorless liquid
19 Aug 2014	2 mg/mL	0.31 mL	4.69 mL	Clear, colorless liquid
	6 mg/mL	0.93 mL	4.07 mL	Clear, colorless liquid
	10 mg/mL	1.58 mL	3.42 mL	Clear, colorless liquid
21 Aug 2014	2 mg/mL	0.31 mL	4.69 mL	Clear, colorless liquid
	6 mg/mL	0.93 mL	4.07 mL	Clear, colorless liquid
	10 mg/mL	1.58 mL	3.42 mL	Clear, colorless liquid
22 Aug 2014	2 mg/mL	0.31 mL	4.69 mL	Clear, colorless liquid
	6 mg/mL	0.93 mL	4.07 mL	Clear, colorless liquid
	10 mg/mL	1.58 mL	3.42 mL	Clear, colorless liquid
25 Aug 2014	2 mg/mL	0.31 mL	4.69 mL	Clear, colorless liquid
	6 mg/mL	0.93 mL	4.07 mL	Clear, colorless liquid
	10 mg/mL	1.58 mL	3.42 mL	Clear, colorless liquid

The appropriate amount of LT3114 stock solution was measured using appropriately sized sterile syringes and dispensed into appropriately sized sterile, clear glass vials. Vehicle was measured using appropriately sized sterile syringes and added to the vial. The vial was gently swirled until the material appeared homogeneous. Formulation samples were taken on the first day of dosing. The vial was sealed with a sterile rubber stopper and crimp cap and wrapped in aluminum foil to protect from light and kept at room temperature until used for dosing. pH was taken on the first day of dosing using Litmus paper. The pH for the vehicle on the first day of dosing was 5.

Before dosing, visual inspection of the formulation was conducted against a black background for any irregularities.

4. *Dose Formulation Samples*

On the first day of dosing, duplicate 1.0 mL samples were collected from the top, middle and bottom of each formulation to confirm concentration and homogeneity. Duplicate samples (1.0 mL) from the middle of the vehicle were collected to confirm the absence of test article. These samples were stored at 2-5°C until shipment for dose formulation analysis.

5. *Dose Formulation Analysis*

Dose formulation samples were shipped to (on wet ice) and analyses were performed under the direction of:

Victor J. Johnson, Ph.D.
Grants & Contracts PI/Study Director
Burleson Research Technologies
ATTN: Kristen Fisher
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Fax: (866) 971-2128
Email: vjohnson@brt-labs.com

Good Laboratory Practice (GLP) regulations require that all formulated test articles be appropriately verified for concentration, homogeneity and stability.

Targeted Total Number of

Formulation Samples: 20 Collected, 10 initial shipment (one set shipped for analysis and the second set retained at Calvert as a backup)

Sample Storage:

Conditions: Refrigerated (2-8°C)

Sample Ship Date: 3 Sep 2014

Testing Requirements: Concentration and homogeneity

Analytical Methods: BRT 20140629-A

Automated Data

Collection Systems: Spectramax 340 microplate spectrophotometer and/or Spectramax M2e microplate spectrophotometer

Analysis of the ELISA absorbance data is performed with Softmax Pro v2.2.1

Analysis of dose formulation acceptance criteria was performed using Microsoft Excel v2013

Dose formulation stability has been conducted under BRT 20140629-A.

If any samples are re-analyzed, the Study Director was notified and details were included in the analytical report.

All Quality Assurance Unit activities and Quality Assurance Unit reports for the analytical analyses were sent to the Calvert Study Director and Calvert Management. All protocol, SOP and GLP deviations associated with the analytical analyses were reported to the Calvert Study Director. The final analytical report includes a GLP compliance statement and a Quality Assurance Unit statement which has the dates of any inspections and the dates the QAU reports were sent to the Principal Investigator, Study Director, and Study Director's Management.

A report of the sample analysis procedures and results was provided to the Study Director and is included in Appendix III. Analyses demonstrated the

absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 2.0 – 2.1 mg/mL, 5.6 – 6.0 mg/mL, and 11.3 – 11.5 mg/mL for the 2.0, 6.0, and 10.0 mg/mL dosing solutions, respectively. The LT3114 concentrations determined for all dose formulation samples met acceptance criteria for accuracy and precision. LT3114 concentrations from the top, middle, and bottom fractions of the dose formulations had a CV% of 2.8%, 3.4%, and 0.9% demonstrating homogeneity of the 2.0, 6.0, and 10.0 mg/mL dose formulations

Remaining samples were appropriately discarded prior to finalization of the study report.

6. Sample Batch Acceptance Criteria

For a sample batch run to be acceptable, the following assay acceptance criteria were met;

Calibration Curve Acceptance Criteria:

- Accuracy of standards: 20% of nominal (25% near LLOQ)
- Precision of the standards: %CV value $\leq 20.0\%$
- At least 75% of the standards (minimum of 6 points) are required to meet these criteria. Failed standards were removed from the analysis.

QC Sample Acceptance Criteria:

- Accuracy of QC samples: 20% of nominal (25% near LLOQ)
- Precision of the QCs: %CV value $\leq 20.0\%$
- At least 67% of all QC samples are required to meet these criteria. In addition, at least 50% of the QC samples at each level are required to meet these criteria.

Concentration and homogeneity of dosing formulations was determined by assay of formulation aliquots sampled from the top, middle and bottom locations of the dosing formulation container. Measured concentration of the dosing formulation was acceptable if it was 80% to 120% of the nominal concentration. Formulated LT3114 was considered homogeneous if the measured concentrations of the top, middle and bottom aliquots had a %CV value $\leq 20.0\%$.

7. Reserve Archive Samples

Since the study was of short duration, no retention sample of the test article was maintained at Calvert.

8. Accountability and Disposition

Unused test article was retained for use in related studies.

B. Test System (Animals and Animal Care)

1. Description

Species:	Rat
Stock:	Hsd:SD
Total Number:	Twenty four (6/group)
Gender:	Male
Age Range:	8-10 weeks
Body Weight Range:	275-319 grams
Animal Source:	Harlan Laboratories
Experimental History:	Purpose-bred and experimentally naïve
Identification:	Ear tagged and color coded

2. Rationale for Choice of Species and Number of Animals

The rat was selected because the physiologic response of their airways is similar to that of a human (1, 2, 3). Previous research has been conducted with the test article in rats.

Based on Calvert's experience with this model and available literature on pulmonary effects of test drugs (4, 5, 6), the number of animals used in this study is the minimum number considered to provide valid scientific assessment.

3. Husbandry

Housing: Animals were group housed in compliance with the National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals were kept was documented in

the study records. No other species were kept in the same room. The experiment was conducted in a room dedicated to this study and the dosing occurred in this room only.

Lighting:	12 hours light/12 hours dark
Room Temperature:	19 to 23°C
Relative Humidity:	38 to 76%
Food:	All animals had access to Certified Rodent Diet (Teklad) equivalent <i>ad libitum</i> , except during dosing and respiratory evaluations. The lot number and specifications of each lot used are archived at Calvert. No contaminants were known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which are maintained in the Calvert archives and a copy is included in the final report.
Water:	Water was available <i>ad libitum</i> , except during dosing and respiratory evaluations, to each animal via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert SOP's. No contaminants are known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis are maintained in the Calvert archives and a copy is included in the final report.
Acclimation:	Study animals were acclimated to their housing for 7-19 days prior to dosing.

4. Pre-study Health Screen and Selection Criteria

All animals received for this study were assessed as to their general health by a suitably trained authorized person. During the acclimation period, each rat was observed at least once daily for any abnormalities or for the

rat was observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were determined to be suitable for use were assigned to this study.

5. *Assignment to Study Groups*

Animals were selected by body weight and apparent good health, and were randomly assigned to study groups. Rats were given the following identification numbers:

Group Number	Color Code	Males
1	Green	1841-1846
2	Blue	1847-1852
3	Yellow	1853-1858
4	Red	1859-1864

6. *Humane Care of Animals*

Treatment of animals was in accordance with the study protocol and also in accordance with Calvert SOP's which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, 2011, National Academies Press). The Calvert IACUC approved the study protocol prior to finalization.

C. *Test Article Administration*

1. *Group Assignments and Dose Levels*

Group	Treatment	Dose (mg/kg)	Volume (mL/kg)	Concentration (mg/mL)	No. of Animals
					Male
1	Diluent	0	5	0	6
2	LT3114	10	5	2.0	6
3	LT3114	30	5	6	6
4	LT3114	50	5	10	6

2. *Dosing*

Route: Slow intravenous injection (~1 minute)

Frequency: Once

Procedure: Doses were administered by intravenous injection via the lateral tail vein at a dose volume of 5 ml/kg.

Each intravenous dose was administered as a manual slow injection (~1 minute). Each animal received a mg/kg dose based on its most recent body weight.

Due to the number of respiratory evaluations and the number of plethysmographs available, only 4 animals could be tested each day. To run 24 rats, testing was conducted over 6 days. Each day one animal from each dose group was tested. The days of dosing were as follows:

Date of Testing	Rat Numbers Evaluated
13 Aug 2014	1841, 1847, 1853, 1859
14 Aug 2014	1842, 1848, 1854, 1860
19 Aug 2014	1843, 1849, 1855, 1861
21 Aug 2014	1844, 1850, 1856, 1862
22 Aug 2014	1845, 1851, 1857, 1863
25 Aug 2014	1846, 1852, 1858, 1864

3. *Justification for Route and Dose Levels*

The intravenous route was chosen as it is the intended route of administration in humans.

The dose levels were selected by the Sponsor based on in-life study results of a 14-day repeat intravenous dose range finding toxicity study of LT3114 in Sprague Dawley Rats. (Calvert study 0437RL35.002).

D. Method of Study Performance

Rats were trained on two days in the head-out plethysmographic chamber for approximately 15 minutes each, before the day of the experiment. On the day of dosing, each animal was weighed and placed in its plethysmographic chamber and baseline values were obtained for 5 minutes following a minimum 5-minute stabilization period. The rat was then removed from the chamber and dosed as per group assignments. Following dosing, each animal was returned to its designated plethysmograph chamber and the respiratory parameters were measured 15 minutes (± 3 min) and 1, 2 and 4 hours (± 5 minutes) following administration of test article/vehicle. Animals were allowed to stabilize in the

minute recording periods began at the designated time following dosing. The following parameters were recorded using the DSI Dataquest Open ART (v. 2.3) & PONEMAH Physiology Platform (v 4.2.0):

- Tidal Volume
- Minute Volume
- Respiratory rate

E. Terminal Procedures

1. Termination

a) Scheduled Sacrifice

All animals were euthanized by CO₂ asphyxiation following the 4 hour data collection. No necropsies were performed, no tissues were saved and the carcasses were discarded.

VIII. Records and Reports

A. Data Collection and Statistical Analysis

Respiratory Data was acquired and analyzed using DSI Dataquest Open ART (v. 2.3) & PONEMAH Physiology Platform (v. 4.2.0). Individual values of respiratory rate, tidal volume and minute volume for the test article treated group were compared back to the vehicle control and baseline using a two-way repeated measures ANOVA followed by a Bonferroni Multiple Comparison Test (SigmaStat, v. 2.03). Differences with p values ≤ 0.05 were considered statistically significant. All other data was hand recorded and manually entered into the report as applicable.

B. Storage of Records

Test article preparation, test article tracking, in-life data, protocol, protocol amendments and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

All dose formulation data will be archived at Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560.

IX. Results

A. Respiratory Parameters

Effects of the vehicle and 10, 30 and 50 mg/kg doses of LT3114 are shown in Table 1.

1. Effects of Vehicle

The intravenous administration of the vehicle, LT3114 Diluent, at 5 mL/kg did not produce any statistically significant changes in respiratory rate or tidal volume compared to baseline values in male rats.

A statistically significant ($p \leq 0.05$) decrease (14%) in minute volume was observed at 1 hour following dosing of the vehicle when compared to the predose (baseline) values. Sporadic changes like this are occasionally observed in this model and not considered biologically relevant.

2. Effects of LT3114

As presented in Table 1, the intravenous administration of LT3114 at 10 mg/kg produced a statistically significant ($p \leq 0.05$) decrease (19%) in respiratory rate at 4 hours following dosing when compared to pre-dose values. The decrease in respiratory rate was not statistically significant when compared to the vehicle control group. The decrease in respiratory rate observed in the 10 mg/kg dose group at 4 hours post-dose was not dose-dependent at the higher doses and therefore not considered test article related. Minute volume and tidal volume were not affected by the intravenous administration of LT3114 at 10 mg/kg.

Statistically significant ($p \leq 0.05$) decreases (8%) in minute volume were observed at 4 hours following the intravenous administration of LT3114 at 30 and 50 mg/kg compared to the vehicle control group values. The changes were not statistically significant compared to the baseline values. Also the predose minute volume values were somewhat higher in the vehicle group than in the test article groups especially in Groups 3 and 4. Since the magnitude of the changes in minute volume was similar to the magnitude of changes occasionally observed in the vehicle control group, these changes were not considered biologically relevant. Respiratory rate and tidal volume were not affected by the 30 or 50 mg/kg doses of LT3114.

Individual values and percent changes are presented in Appendix I.

X. Conclusion

The intravenous administration of LT3114 at 10, 30 and 50 mg/kg did not induce any biologically relevant effects on respiratory rate, tidal volume or minute volume in conscious male rats.

XI. References

1. ICH S7A Safety Pharmacology Studies for Human Pharmaceuticals, July, 2001.
2. Peiser C, Udem BJ, Fischer A. Nociceptin effects in the airways. Peptides., 2000 Jul;21(7):995-8.
3. Murphy, D.J. Assessment of respiratory function in safety pharmacology. Fundam. Clin. Pharmacol. 16: 183-196, 2002.
4. Schlenker, E.H., Shi, Y., Wipf, J., Martis, D.S., Kost, C.K. Fructose feeding and intermittent hypoxia affect ventilatory responsiveness to hypoxia and hypercapnia in rats. J. Appl. Physiol. 97: 1387-1394, 2004
5. Kaczynska, K. and Szereda-Przestaszewska, M. Clonidine-induced respiratory effects in anesthetized rats. Exp. Physiol. 91: 269-275 (2007).
6. Bailey, T.C., Maruscak, A.A., Petersen, A., White, S., Lewis, J.F., Veldhuizen, A.W. Physiological effects of oxidized exogenous surfactant in vivo: effects of high tidal volume and surfactant protein A. Am. J. Physiol. Lung Cell. Mol. Physiol. 291: L703-L709 (2006).

XII. Table

A. Table 1- Evaluation of Respiratory Function of LT3114 in Male Rats (n=6)

Intravenous Treatment ^a	Time (min)	Respiratory Rate (breaths/min)			Tidal Volume (mL)			Minute Volume (mL/min)		
		Mean	%Δ	sem	Mean	%Δ	sem	Mean	%Δ	sem
Vehicle control LT3114 Diluent 5 mL/kg	0	172.63	0	3.10	1.11	0	0.03	184.77	0	7.04
	15	180.52	5	12.39	1.02	-9	0.04	173.53	-6	5.40
	60	180.22	4	13.57	0.93	-16	0.04	159.63*	-14	4.66
	120	178.73	4	9.69	0.98	-12	0.05	168.46	-9	5.15
	240	178.58	3	13.85	1.09	-3	0.05	183.48	-1	4.96
Intravenous Treatment ^a	Time (min)	Respiratory Rate (breaths/min)			Tidal Volume (mL)			Minute Volume (mL/min)		
		Mean	%Δ	sem	Mean	%Δ	sem	Mean	%Δ	sem
LT3114 10 mg/kg 5 mL/kg 2 mg/mL	0	185.69	0	11.74	1.00	0	0.08	175.07	0	8.98
	15	173.76	-6	8.12	1.08	8	0.06	178.17	2	4.38
	60	182.22	-2	11.70	0.95	-5	0.05	165.68	-5	7.57
	120	175.30	-6	11.99	1.00	0	0.05	166.32	-5	6.13
	240	150.06*	-19	4.71	1.16	16	0.06	171.13	-2	10.14
Intravenous Treatment ^a	Time (min)	Respiratory Rate (breaths/min)			Tidal Volume (mL)			Minute Volume (mL/min)		
		Mean	%Δ	sem	Mean	%Δ	sem	Mean	%Δ	sem
LT3114 30 mg/kg 5 mL/kg 6 mg/mL	0	168.97	0	7.88	1.07	0	0.05	169.75	0	6.37
	15	182.33	8	12.61	0.92	-14	0.06	156.48	-8	5.61
	60	176.85	5	13.39	0.92	-14	0.06	152.52	-10	3.94
	120	190.88	13	16.69	0.91	-15	0.06	160.75	-5	6.40
	240	150.63	-11	12.84	1.10	3	0.05	156.54**	-8	6.78
Intravenous Treatment ^a	Time (min)	Respiratory Rate (breaths/min)			Tidal Volume (mL)			Minute Volume (mL/min)		
		Mean	%Δ	sem	Mean	%Δ	sem	Mean	%Δ	sem
LT3114 50 mg/kg 5 mL/kg 10 mg/mL	0	167.68	0	6.74	1.03	0	0.04	165.31	0	3.94
	15	166.83	-1	7.65	1.00	-4	0.03	160.70	-3	6.19
	60	170.86	2	8.08	0.97	-7	0.05	158.54	-4	3.45
	120	179.06	7	12.78	0.96	-7	0.05	162.51	-2	3.93
	240	154.37	-8	14.97	1.04	1	0.04	152.72**	-8	8.36

Mean and standard error of the mean calculated using non-truncated values.

^a Doses were administered as a slow intravenous injection (~1 minute)

* = Statistically significant ($p \leq 0.05$) change from time 0

** = Statistically significant ($p \leq 0.05$) change from vehicle control group

XIII. Appendices

A. Appendix I - Individual Data

Individual Ventilatory Parameters – Male Rats

Intravenous Treatment	Time (Hours)	Respiratory Rate (breaths/min)											
		1841	%Δ	1842	%Δ	1843	%Δ	1844	%Δ	1845	%Δ	1846	%Δ
Group 1	0	178.51	0	158.84	0	171.36	0	178.97	0	177.09	0	170.99	0
Vehicle control	0.25	161.88	-9	143.39	-10	191.66	12	206.97	16	220.35	24	158.89	-7
Diluent	1	158.39	-11	166.79	5	200.02	17	237.29	33	172.74	-2	146.12	-15
0 mg/kg	2	175.68	-2	162.22	2	183.95	7	221.99	24	174.71	-1	153.84	-10
5 mL/kg	4	142.21	-20	169.39	7	181.11	6	242.55	36	164.01	-7	172.22	1
0 mg/mL													

Intravenous Treatment	Time (Hours)	Respiratory Rate (breaths/min)											
		1847	%Δ	1848	%Δ	1849	%Δ	1850	%Δ	1851	%Δ	1852	%Δ
Group 2	0	158.62	0	175.87	0	203.91	0	212.21	0	215.26	0	148.29	0
LT3114	0.25	172.34	9	162.53	-8	201.68	-1	164.70	-22	192.71	-10	148.63	0
10 mg/kg	1	198.54	25	148.62	-15	177.72	-13	204.44	-4	215.16	0	148.82	0
5 mL/kg	2	157.54	-1	160.37	-9	177.01	-13	202.36	-5	216.17	0	138.33	-7
2 mg/mL	4	147.83	-7	141.90	-19	170.49	-16	145.06	-32	155.86	-28	139.25	-6

Intravenous Treatment	Time (Hours)	Respiratory Rate (breaths/min)											
		1853	%Δ	1854	%Δ	1855	%Δ	1856	%Δ	1857	%Δ	1858	%Δ
Group 3	0	134.49	0	185.13	0	186.63	0	176.60	0	168.70	0	162.27	0
LT3114	0.25	138.69	3	159.84	-14	226.04	21	203.12	15	186.10	10	180.17	11
30 mg/kg	1	141.03	5	161.46	-13	180.02	-4	175.43	-1	237.79	41	165.38	2
5 mL/kg	2	125.85	-6	157.48	-15	196.59	5	229.45	30	221.62	31	214.31	32
6 mg/mL	4	130.86	-3	134.64	-27	149.98	-20	194.97	10	180.54	7	112.75	-31

Individual Ventilatory Parameters- Male Rats

Intravenous Treatment	Time (Hours)	Respiratory Rate (breaths/min)											
		1859	%Δ	1860	%Δ	1861	%Δ	1862	%Δ	1863	%Δ	1864	%Δ
Group 4 LT3114 50 mg/kg 5 mL/kg 10 mg/mL	0	148.89	0	190.18	0	159.29	0	185.51	0	164.81	0	157.39	0
	0.25	187.32	26	166.98	-12	146.90	-8	175.16	-6	182.66	11	141.97	-10
	1	155.80	5	153.46	-19	153.30	-4	188.06	1	175.34	6	199.19	27
	2	150.57	1	168.20	-12	153.86	-3	235.45	27	190.92	16	175.36	11
	4	136.66	-8	133.57	-30	118.63	-26	213.18	15	185.83	13	138.37	-12

Individual Ventilatory Parameters – Male Rats

Intravenous Treatment	Time (Hours)	Tidal Volume (mL)											
		1841	%Δ	1842	%Δ	1843	%Δ	1844	%Δ	1845	%Δ	1846	%Δ
Group 1	0	1.02	0	1.05	0	1.10	0	1.16	0	1.23	0	1.12	0
Vehicle control	0.25	1.04	2	1.10	5	1.01	-8	0.94	-19	0.88	-28	1.12	0
Diluent	1	1.00	-2	0.91	-13	0.88	-20	0.75	-35	0.99	-20	1.05	-6
0 mg/kg	2	1.03	1	0.89	-15	0.96	-13	0.82	-29	1.02	-17	1.17	4
5 mL/kg	4	1.21	18	1.06	1	1.03	-7	0.87	-25	1.24	0	1.10	-1
0 mg/mL													

Intravenous Treatment	Time (Hours)	Tidal Volume (mL)											
		1847	%Δ	1848	%Δ	1849	%Δ	1850	%Δ	1851	%Δ	1852	%Δ
Group 2	0	1.23	0	0.81	0	0.87	0	0.86	0	0.94	0	1.29	0
LT3114	0.25	1.09	-11	1.07	33	0.85	-3	1.23	43	1.00	6	1.25	-3
10 mg/kg	1	0.94	-23	0.95	17	0.87	-1	0.93	8	0.83	-12	1.21	-6
5 mL/kg	2	1.07	-13	1.01	25	0.89	2	0.93	8	0.91	-4	1.19	-8
2 mg/mL	4	1.21	-2	1.17	44	1.05	20	1.32	54	1.29	37	0.90	-30

Intravenous Treatment	Time (Hours)	Tidal Volume (mL)											
		1853	%Δ	1854	%Δ	1855	%Δ	1856	%Δ	1857	%Δ	1858	%Δ
Group 3	0	1.14	0	0.93	0	0.93	0	1.09	0	1.13	0	1.21	0
LT3114	0.25	1.14	0	0.87	-6	0.67	-27	0.88	-20	0.95	-16	1.02	-16
30 mg/kg	1	1.07	-6	0.90	-4	0.86	-7	1.00	-8	0.66	-41	1.06	-12
5 mL/kg	2	1.16	2	0.95	3	0.86	-7	0.74	-32	0.79	-30	0.94	-22
6 mg/mL	4	1.13	-1	1.23	32	1.09	18	1.01	-8	0.95	-16	1.21	1

Individual Ventilatory Parameters - Male Rats

Intravenous Treatment	Time (Hours)	Tidal Volume (mL)											
		1859	%Δ	1860	%Δ	1861	%Δ	1862	%Δ	1863	%Δ	1864	%Δ
Group 4 LT3114 50 mg/kg 5 mL/kg 10 mg/mL	0	1.11	0	0.86	0	1.00	0	1.01	0	1.08	0	1.15	0
	0.25	0.92	-17	1.09	26	1.07	8	1.00	-1	0.93	-14	0.96	-16
	1	1.03	-7	1.14	32	0.99	-1	0.87	-14	0.94	-13	0.82	-29
	2	1.07	-3	1.10	28	1.00	0	0.75	-26	0.89	-18	0.95	-17
	4	1.12	1	1.18	36	1.05	5	0.90	-11	0.93	-13	1.07	-7

Individual Ventilatory Parameters – Male Rats

Intravenous Treatment	Time (Hours)	Minute Volume (mL)											
		1841	%Δ	1842	%Δ	1843	%Δ	1844	%Δ	1845	%Δ	1846	%Δ
Group 1	0	176.37	0	159.88	0	187.15	0	194.67	0	210.66	0	179.86	0
Vehicle control	0.25	166.99	-5	156.05	-2	181.84	-3	188.71	-3	184.44	-12	163.17	-9
Diluent	1	153.25	-13	148.34	-7	171.28	-8	170.41	-12	167.70	-20	146.78	-18
0 mg/kg	2	171.97	-2	143.31	-10	174.46	-7	176.35	-9	175.71	-17	168.95	-6
5 mL/kg	4	169.21	-4	175.10	10	183.49	-2	202.34	4	192.49	-9	178.25	-1
0 mg/mL													

Intravenous Treatment	Time (Hours)	Minute Volume (mL)											
		1847	%Δ	1848	%Δ	1849	%Δ	1850	%Δ	1851	%Δ	1852	%Δ
Group 2	0	193.56	0	135.82	0	168.99	0	174.81	0	196.83	0	180.39	0
LT3114	0.25	176.64	-9	165.24	22	166.84	-1	192.47	10	185.88	-6	181.94	1
10 mg/kg	1	174.85	-10	136.69	1	149.11	-12	184.98	6	173.61	-12	174.84	-3
5 mL/kg	2	160.46	-17	153.07	13	155.05	-8	181.10	4	189.13	-4	159.13	-12
2 mg/mL	4	175.25	-9	163.16	20	172.14	2	189.55	8	199.01	1	127.70	-29

Intravenous Treatment	Time (Hours)	Minute Volume (mL)											
		1853	%Δ	1854	%Δ	1855	%Δ	1856	%Δ	1857	%Δ	1858	%Δ
Group 3	0	144.30	0	163.22	0	163.67	0	181.38	0	181.22	0	184.70	0
LT3114	0.25	151.75	5	137.63	-16	145.91	-11	161.30	-11	171.33	-5	170.95	-7
30 mg/kg	1	146.67	2	141.86	-13	149.93	-8	158.67	-13	149.25	-18	168.73	-9
5 mL/kg	2	144.75	0	143.88	-12	160.15	-2	161.09	-11	169.00	-7	185.62	0
6 mg/mL	4	140.95	-2	162.78	0	158.90	-3	177.02	-2	166.83	-8	132.74	-28

Individual Ventilatory Parameters – Male Rats

Intravenous Treatment	Time (Hours)	Minute Volume (mL)											
		1859	%Δ	1860	%Δ	1861	%Δ	1862	%Δ	1863	%Δ	1864	%Δ
Group 4 LT3114 50 mg/kg 5 mL/kg 10 mg/mL	0	157.63	0	156.92	0	155.90	0	176.65	0	169.28	0	175.48	0
	0.25	161.66	3	178.65	14	154.98	-1	170.45	-4	163.95	-3	134.50	-23
	1	156.38	-1	173.33	10	147.30	-6	157.52	-11	159.98	-5	156.73	-11
	2	154.69	-2	178.05	13	150.46	-3	166.19	-6	163.87	-3	161.82	-8
	4	148.24	-6	155.63	-1	121.75	-22	183.29	4	163.00	-4	144.42	-18

B. Appendix II - Test Article Information

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
			PAGE

Name		Part Number		Lot Number
Lpathomab-DS, 30 mg/mL		TBD		GA121459
Formulation Buffer				Storage Temperature
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0				2 to 8°C
Production Date		Test Date	Retest Date	Intended Use
JUN 2014		JUL 2014	DEC 2014	GLP-conforming Toxicity Testing
Manufacturer		Manufacturer Part Number		Manufacturer Lot Number
Gallus-MO, St. Louis, MO		TBD		Gallus-MO 103948
Attribute	Test	Specification		Results
Identity	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution		Particulate free, clear, slightly yellow solution
	pH	5.0 ± 0.2		5.1
	CGE (Non-reduced)	Major band at 150 kDa		Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa		Conforms
	iCE	Comparable to RS		Conforms
		Minor Peak at pI ~8.3 Major Peak at pI ~8.4 Minor Peak at pI ~8.6 Faint Peak at pI ~8.7		15% 60% 13% 4%
Potency	API Concentration by OD at 280 nm	30 ± 3 mg/mL		32.1 mg/mL
	LPA Binding	100% ±40% of RS		89%

Confidential

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
		PAGE	2 of 2

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected

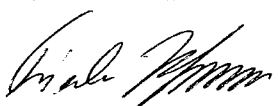
Attribute	Test	Specification	Results
Safety	Residual Protein-A	≤ 10 ppm	<1 ppm
	Residual DNA	≤ 5 pg/mg	<0.05 pg/mg
	CHO HCP	≤ 100 ppm	15 ppm
	Endotoxin, USP <85>	Report	<0.06 EU/mL ¹
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL	None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

Quality Assurance:


Acting Head of QA - Frieder K. Hofmann, PhD

Date: 22AUG2014

¹ At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

Lpath Certificate of Analysis							
Stability Assessment of Lpathomab-DS, lot 103948, stored at 5 ± 3°C			Formulation Age				
Assay	Specification	Time 0	1 Month	2 Months	3 Months		
Appearance	Particulate-free, clear, colorless to slightly yellowish and slightly opalescent solution.	Conforms	Conforms	Conforms	Conforms		
pH	4.8 to 5.2	5.2	5.1	5.1	5.1		
ICE	Comparable to Reference Standard	Conforms	Conforms	Conforms	Conforms		
	Report pI and relative peak area of Major Isoform and total relative peak area of acidic and basic variants	pI	Area	pI	Area	pI	Area
		Acidic	23%	Acidic	22%	Acidic	23%
		8.2	53%	8.2	53%	8.2	52%
		Basic	24%	Basic	25%	Basic	25%
CGE-NR	IgG + IgG-NG ≥ 90%	98.4%	97.9%	98.8%	97.6%		
	IgG-NG	2.1%	1.8%	1.5%	2.2%		
	2H-1L	0.3%	0.4%	0.4%	0.4%		
	2H	0%	0%	0%	0%		
	HL	0%	0%	0%	0%		
	HC	0%	0%	0%	0%		
	LC	0.4%	0.4%	0.5%	0.6%		
CGE-R	HC + HC-NG + LC ≥ 90%	98.1%	95.5%	99.1%	98.1%		
	HC	66%	63%	67%	66%		
	HC-NG	1.1%	1.5%	1.1%	1.1%		
	LC	31%	31%	31%	31%		
qSE-HPLC	API concentration: 29 to 35 mg/mL	33.2	33.5%	34.3	34.5		
	Purity: ≥ 90%	99.2	98.6%	98.3%	98.2		
	Aggregates: ≥ 5%	0.8%	1.4%	1.7%	1.8		
	No single impurity > %	0%	0%	0%	0%		
LPA-binding ELISA	60% to 140% of reference standard activity	125%	106%	92%	136%		
Particulate	≤ 6,000 Particles ≥ 10 µm per vial	51			51		
	≤ 600 Particles ≥ 25 µm per vial	2			0		
Judgment							
Lot 103948 of Lpathomab-DS was placed in a validated 5 ± 3°C stability chamber at Lpath's qualified contract laboratory, Eurofins Lancaster Laboratories in Lancaster, PA, and tested monthly for three months with the stability-indicating assays appearance, pH, isoelectric focusing by capillary electrophoresis (ICE), non-reduced and reduced capillary gel electrophoresis (CGE-NR and CGE-R), quantitative size-exclusion high performance liquid chromatography (qSE-HPLC), LPA-binding ELISA and subvisible particulate matter. The obtained results and underlying raw data were reviewed and verified by Lpath's Quality Department. The results only varied within each test-specific accuracy tolerance and indicate that the lot exhibited excellent product stability under the storage conditions.							
Department	Title	Name	Signature	Date			
Quality Assurance	Head of Product Quality	Frieder K Hofmann, PhD		22DEC2014			

C. Appendix III - Test Article Formulation Analyses

STUDY PHASE REPORT

STUDY TITLE: Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)

REPORT TITLE: Determination of LT3114 in Dose Formulation Samples

BRT Reference No.: BRT 20140814

Author: Victor J. Johnson, PhD (BRT)

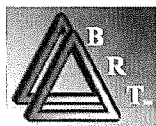
**Dosing Formulation
Analyses Laboratory:** Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Number: 1275RL35.001

Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121-1404

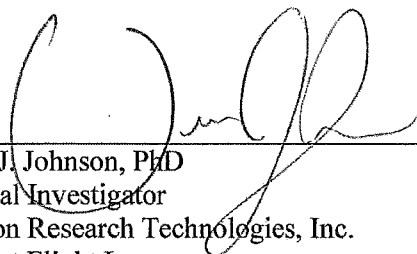
Report Date: 19 December 2014



COMPLIANCE STATEMENT

This study was performed according to the study protocol, applicable BRT SOPs, and in compliance with the following regulations:

- Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58)
- OECD Principles on Good Laboratory Practice (C[97]186/Final)

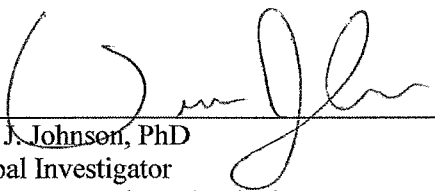


Victor J. Johnson, PhD
Principal Investigator
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
USA

19 Dec 14

Date

APPROVALS



Victor J. Johnson, PhD
Principal Investigator
Burleson Research Technologies, Inc.

19 Dec 14

Date

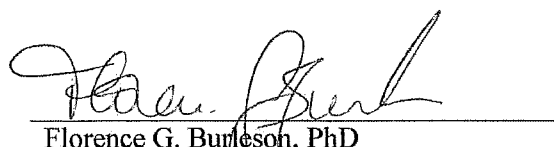


Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.

19 Dec 2014

Date

Report Review:



Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.

19 Dec 14

Date

QUALITY ASSURANCE STATEMENT

Study Phase Reference Number: BRT 20140814

Study Phase Title: Determination of LT3114 in Dose Formulation Samples

Testing Site: Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560

Principal Investigator: Victor J. Johnson, PhD

Phase Inspected	Date of Inspection	Date Report Sent to BRT Principal Investigator/Management	Date Report Sent to Calvert Study Director/Management
Sample Addition Audit	09 Dec 14	09 Dec 14	09 Dec 14
Raw Data Audit	17 Dec 14	17 Dec 14	19 Dec 14
Report Audit	17 Dec 14	17 Dec 14	19 Dec 14

The final report and data record audit were performed by the BRT Quality Assurance Unit.

To the best of my knowledge, this final report accurately describes the study methods and procedures used, and the reported results accurately reflect the raw data.



Christine Ford
Quality Assurance Manager



Date

TABLE OF CONTENTS

COMPLIANCE STATEMENT.....	2
APPROVALS.....	3
QUALITY ASSURANCE STATEMENT	4
TABLE OF CONTENTS	5
STUDY INFORMATION	6
SUMMARY	7
INTRODUCTION.....	8
STUDY DESIGN.....	8
<i>DOSE FORMULATION SAMPLES</i>	<i>8</i>
METHODOLOGY	8
<i>DOSE FORMULATION SAMPLES</i>	<i>8</i>
<i>LT3114 ELISA METHOD.....</i>	<i>9</i>
<i>DATA AND STATISTICAL ANALYSIS.....</i>	<i>9</i>
<i>ACCEPTANCE CRITERIA (PER METHOD VALIDATION BRT 20140629A)</i>	<i>9</i>
RESULTS AND CONCLUSIONS	10
<i>DOSE FORMULATION SAMPLE ANALYSIS.....</i>	<i>10</i>
DOCUMENTATION AND RETENTION OF DATA	11
<i>DESCRIPTION OF CIRCUMSTANCES AFFECTING DATA QUALITY OR INTEGRITY</i>	<i>11</i>
<i>MAINTENANCE OF STUDY RECORDS</i>	<i>11</i>
<i>DEVIATIONS FROM THE STUDY PROTOCOL.....</i>	<i>11</i>
TABLES.....	12
APPENDIX 1: ELISA METHOD FOR QUANTIFICATION OF LT3114.....	14

STUDY INFORMATION

Determination of LT3114 in Dose Formulation Samples

Study Title:	Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)
Testing Facility:	Calvert Laboratories, Inc.
Testing Facility Study No:	1275RL35.001
Study Director:	Brian Klatt, B.S. (Calvert Laboratories)
Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121-1404
Test Site (Dosing Formulation Analyses):	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Test Site Reference No.:	BRT 20140814
Principal Investigator:	Victor J. Johnson, PhD (BRT)
Analyst:	Kristen Fisher (BRT)
Study Initiation Date:	01 Aug 14
Sample Receipt Date:	04 Sep 14
Experimental Start Date:	09 Dec 14
Experimental Completion Date:	12 Dec 14
Study Phase Completion Date:	19 Dec 14

SUMMARY

The purpose of this study phase was to determine the concentration of LT3114 in dose formulation samples generated by Calvert Study No. 1275RL35.001: “Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)”.

Dose formulation samples were provided for three dose concentrations, 2.0 mg/mL, 6.0 mg/mL, and 10.0 mg/mL. Dose formulation samples were collected from each dosing solution and the vehicle, from the top, middle, and bottom strata of the LT3114 dose formulations and from the middle stratum of the vehicle. Analyses demonstrated the absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 2.0 – 2.1 mg/mL, 5.6 – 6.0 mg/mL, and 11.3 – 11.5 mg/mL for the 2.0, 6.0, and 10.0 mg/mL dosing solutions, respectively. The LT3114 concentrations determined for all dose formulation samples met acceptance criteria for accuracy and precision. LT3114 concentrations from the top, middle, and bottom fractions of the dose formulations had a CV% of 2.8%, 3.4%, and 0.9% demonstrating homogeneity of the 2.0, 6.0, and 10.0 mg/mL dose formulations.

INTRODUCTION

The objective of this study phase was to determine the concentration of LT3114 in dose formulation samples in support of the study 1275RL35.001: “Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)”.

STUDY DESIGN

Dose Formulation Samples

Dose formulations (2.0, 6.0, and 10.0 mg/mL) were prepared by Calvert Laboratories on 13 Aug 14 according to the study protocol, using LT3114 in a citrate buffer formulation supplied by the Sponsor. Duplicate 1 mL samples were collected on 13 Aug 14 from the top, middle, and bottom of each dosing solution and from the middle of the vehicle solution. The samples were stored at 2-8°C until single aliquots from each sample were shipped to BRT and received on 04 Sep 14 (Table 1). Dose formulation samples were logged into BRT inventory and stored at 2-8°C until analyzed for LT3114 concentration. Samples were returned to 2-8°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director per protocol. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 1: Dose formulation samples received at BRT on 04 Sep 14.

Date of Collection	Date of Receipt at BRT	Aliquots	LT3114 Concentration (mg/mL)	Location	Volume (mL)
13 Aug 14	04 Sep 14	1	Vehicle	Middle	1
13 Aug 14	04 Sep 14	1	2.0	Top	1
13 Aug 14	04 Sep 14	1	2.0	Middle	1
13 Aug 14	04 Sep 14	1	2.0	Bottom	1
13 Aug 14	04 Sep 14	1	6.0	Top	1
13 Aug 14	04 Sep 14	1	6.0	Middle	1
13 Aug 14	04 Sep 14	1	6.0	Bottom	1
13 Aug 14	04 Sep 14	1	10.0	Top	1
13 Aug 14	04 Sep 14	1	10.0	Middle	1
13 Aug 14	04 Sep 14	1	10.0	Bottom	1

METHODOLOGY

Dose Formulation Samples

Dose formulation samples (single Vehicle sample plus 3 dose levels with 3 samples each from top, middle, and bottom of the preparations = 10 total samples) were received in

good condition on wet ice on 04 Sep 14. Upon receipt at BRT, all samples were stored at 2-8°C. The samples were analyzed for LT3114 concentrations at BRT.

LT3114 ELISA Method

Dose formulation samples were analyzed for LT3114 concentration using a validated ELISA method (BRT 20140629A). The ELISA method is provided in Appendix 1.

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. Standards, quality control (QC) samples, delipidized bovine serum (DBS) assay buffer, and dose formulation samples were added to the coated plates and incubated to allow for capture of the LT3114. A horseradish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Data and Statistical Analysis

Data analysis was performed using SoftMax[®]Pro (Molecular Devices) software. Calibration curves were generated by plotting the mean OD₄₅₀ values (y-axis) versus concentration (x-axis) and fitting the curve using the 4-parameter logistic function. Computer printouts included OD₄₅₀ values (raw data), the calibration curve, the standard deviation, the coefficient of variation between replicates, and the interpolated values of the unknown test samples. Concentrations of LT3114 in test samples were obtained by multiplying the mean interpolated value (result), from duplicate wells, by the sample dilution factor. Results were reported to one decimal place.

Calculations, including means, standard deviations, and coefficients of variation, for determining accuracy and precision were performed using Microsoft Excel 2013.

Acceptance Criteria (per method validation BRT 20140629A)

Calibration Curve Acceptance Criteria:

- Calibration curve range: 3.9 to 500 ng/mL
- Accuracy of standards: 20% of nominal (25% at LLOQ and ULOQ)
- Precision of the standards within 20%

- At least 75% of the standards (minimum of 6 points) met these criteria. Failed standards were removed from the analysis, if present.

QC Sample Acceptance Criteria:

- QC1-300 ng/mL; QC2-150 ng/mL; QC3-5 ng/mL
- Accuracy of QC samples: 20% of nominal (25% at 5 ng/mL and 300 ng/mL)
- Precision of the QCs were within 20% (25% at 5 ng/mL and 300 ng/mL)
- At least 67% of all QC samples met these criteria. In addition, at least 50% of the QC samples at each level met these criteria.
- Total error ($|\%RE| + \%CV$) of each QC was $\leq 30\%$ ($\leq 40\%$ at 5 ng/mL and 300 ng/mL)

Study Dose Formulation Sample Acceptance Criteria:

- Precision of the replicates was within 20% (25% at LLOQ and ULOQ)
- Sample concentrations were acceptable when accuracy was within 20% of nominal

Study Dose Formulation Sample Reanalysis:

- Samples were reanalyzed if the assay is rejected due to failure of the standards or QCs to meet acceptance criteria provided above.
- Samples were reanalyzed if the mean OD450 for the replicates is outside the range of OD450 values for the calibration curve. If applicable, the sample was diluted appropriately and reanalyzed. This did not apply to samples diluted at the MRD that were below the range of the standard curve.
- For samples that were within the range of the standard curve, samples were reanalyzed if the precision of the replicates was not acceptable.

RESULTS AND CONCLUSIONS

Dose Formulation Sample Analysis

LT3114 concentrations for the triplicate samples (top, middle, and bottom) of each dose formulation are provided in Table 2. Analysis confirmed the absence of LT3114 in the vehicle preparations. The concentration of LT3114 in the 2.0 mg/mL dose formulations was 2.0, 2.0, and 2.1 mg/mL for the 2.0 mg/mL dose formulation, 5.6, 5.8, and 6.0 mg/mL for the 6.0 mg/mL dose formulation, and 11.3, 11.4, and 11.5 mg/mL for the 10.0 mg/mL

dose formulation. All dose formulation concentration met acceptance criteria for accuracy and precision. The homogeneity of each dose formulation was assessed by determining the CV% for the top, middle, and bottom samples collected from each formulation. The CV% was 2.8%, 3.4%, and 0.9% for the 2.0, 6.0, and 10.0 mg/mL dose formulations, respectively, demonstrating acceptable homogeneity for all dose formulations.

All dose formulation samples were analyzed within the 2-8°C stability for LT3114 in a citrate buffer formulation established in the method validation (BRT 20140629A).

DOCUMENTATION AND RETENTION OF DATA

Description of Circumstances Affecting Data Quality or Integrity

BRT is not aware of any circumstances surrounding this study phase that would adversely affect data quality or integrity.

Maintenance of Study Records

All raw data, protocol, amendments, study records, and the final study phase report will be archived at the Test Site (BRT) for a period of at least 2 years from the date that the final study phase report is issued unless alternative arrangements are made with the Sponsor. After that time, or as per contract agreement, the Sponsor will be notified and the data and report will be returned to the Sponsor, unless alternative arrangements are made. All raw data not specific to this study (e.g., instrument logs, CVs, etc.) will be archived at BRT.

Deviations from the Study Protocol

- There was a technical error in preparation of the calibration curve for the assay run performed on 09 Dec 14 that resulted in failure of the QC samples to meet acceptance criteria. The assay run was rejected and the analysis was repeated. There was no impact of these deviation on the study phase since the analysis was successfully repeated.

TABLES

Table 1: Dose formulation samples received at BRT on 04 Sep 14.	8
Table 2: LT3114 concentrations in dose formulation samples.	13

Table 2: LT3114 concentrations in dose formulation samples.

Sampling Dates	Dose Formulation	Location	LT3114 (mg/mL)	% of Target	CV%
13 Aug 14	Vehicle	Middle	BLQ	N/A	N/A
	LT3114 Low-Dose: 2.0 mg/ml	Top	2.1	105	2.8
	LT3114 Low-Dose: 2.0 mg/ml	Middle	2.0	100	
	LT3114 Low-Dose: 2.0 mg/ml	Bottom	2.0	100	
	LT3114 Mid-Dose: 6.0 mg/ml	Top	5.8	97	3.4
	LT3114 Mid-Dose: 6.0 mg/ml	Middle	5.6	93	
	LT3114 Mid-Dose: 6.0 mg/ml	Bottom	6.0	100	
	LT3114 High-Dose: 10.0 mg/ml	Top	11.4	113	0.9
	LT3114 High-Dose: 10.0 mg/ml	Middle	11.3	114	
	LT3114 High-Dose: 10.0 mg/ml	Bottom	11.5	115	

BLQ – Below Lower Limit of Quantification (3.9 ng/mL); N/A – Not applicable.

APPENDIX 1: ELISA METHOD FOR QUANTIFICATION OF LT3114.

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Citrate buffer provided by Lpath
- LT3114 Citrate Formulated Drug Substance (Provided by Sponsor)
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS), assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H₂SO₄ (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)
 - 1.1 LPA-Conjugated-BSA coating material was kept frozen until use.
 - 1.2 After dilution, LPA-Conjugated-BSA coating material was stored at 2-8° C for up to 4 months.
2. Peroxidase conjugated-anti-human IgG secondary antibody
 - 2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%).
 - 2.2 Store at -20° C for up to 1 year.
3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)
 - 3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly.
 - 3.2 Store at 2-8°C for up to 3 months.
4. Blocking Solution 1xPBS, 1% BSA I, 0.1% Tween 20
 - 4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X).
 - 4.2 Mix thoroughly and sterile filter.
 - 4.3 Store at 2-8°C for up to 1 month.

Preparation of Quality Control Samples

Prepare QC samples in citrate buffer using Human Reference Anti-LPA IgG LT3114 at 300, 150, and 10 ng/mL. Aliquot and store at $\leq -70^{\circ}\text{C}$.

ELISA Method

1. Coating Plates
 - 1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at $\leq -70^{\circ}\text{C}$. Thaw the aliquot at room temperature.
 - 1.2 Dilute LPA-Conjugated-BSA coating material to 2.5 $\mu\text{g/ml}$ in 0.05 M Carbonate Buffer.
 - 1.3 Coat a 96-well high-binding microtiter plate(s) with 100 μL /well coating solution.
 - 1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.
2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).
3. Blocking Plates
 - 3.1 Wash coated plate(s) 4 times with wash solution.

- 3.2 Dispense 150 µL/well of blocking solution to each well and cover the plate(s) with plate sealers.
 - 3.3 Incubate plate(s) at 37°C for 1 hour ± 5 minutes.
4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)
 - 4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)
 - 4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.
 - 4.3 Prepare Human Reference Anti-LPA IgG 8 point standard curve by diluting in DBS to achieve desired range (3.9 – 500 ng/ml).
 - 4.4 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 4.5 Add samples (controls, calibrators and unknowns) at 100 µL/well.
5. Add samples to appropriate wells
 - 5.1 Dispense 100 µl of appropriate sample to designated wells in duplicate. Add 100 µl of DBS to assay blank wells.
 - 5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
6. Add HRP-goat-anti-human secondary antibody solution.
 - 6.1 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody
 - 6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.
 - 6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.
 - 6.3 Dispense 100 µl of diluted HRP-goat-anti-human secondary antibody to each well.
 - 6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
7. Substrate Addition
 - 7.1 Wash coated plate(s) 4 times with wash solution.
 - 7.2 Dispense 100 µl **COLD** TMB solution to each well.
 - 7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.

8. Stop Solution Addition

8.1 Dispense 100 μ l 1.0 M H₂SO₄ solution to each well.

8.2 Measure absorbance at 450 nm on microplate reader.

D. Appendix IV – Food and Water Analysis

Laboratory Diet Certification Report

Teklad Certified Global 16% Protein Rodent Diet

EXACT COPY
INITIAL: BIC
DATE: 30 OCT 14

2016C



Lot Number 2016C-062014MA
Date of Manufacture 06/20/14
Report Date 07/09/14

The following data is a consolidation of results obtained from one or more independent testing laboratories. The actual laboratory results are available upon request.

Kurt Schaefer

Quality Assurance Coordinator, Teklad Diets
Research Models and Services
Harlan Laboratories, Inc.

I have reviewed this document
2014.07.10 06:29:50 -05'00'

Proximate Analysis	
Analysis	Result (%)
Protein	16.60
Fat	3.59
Fiber	3.09
Moisture	11.66
Ash	4.98
Calcium	1.00
Phosphorus	0.71

Feed Contaminant Screen			
Analysis	Result	Units	Established Maximum Concentration

Heavy Metals

Arsenic	0.19	ppm	1.00
Cadmium	< 0.10	ppm	0.50
Lead	< 0.20	ppm	1.50
Mercury	< 0.05	ppm	0.20
Selenium	0.28	ppm	0.50

Mycotoxin

Aflatoxin B1, B2, G1, G2	< 5.00	ppb	5.00
--------------------------	--------	-----	------

Chlorinated Hydrocarbons

Aldrin	< 0.01	ppm	0.03
Lindane	< 0.01	ppm	0.05
Chlordane	< 0.01	ppm	0.05
DDT & related substances	< 0.03	ppm	0.15
Dieldrin	< 0.02	ppm	0.03
Endrin	< 0.02	ppm	0.03
Heptachlor	< 0.01	ppm	0.03
Heptachlor Epoxide	< 0.01	ppm	0.03
Toxaphene	< 0.10	ppm	0.15
PCB's	< 0.10	ppm	0.15
a-BHC	< 0.01	ppm	0.05
b-BHC	< 0.01	ppm	0.05
d-BHC	< 0.01	ppm	0.05
Hexachlorobenzene	< 0.01	ppm	0.03
Mirex	< 0.01	ppm	0.02
Methoxychlor	< 0.05	ppm	0.50

Organophosphates

Thimet	< 0.15	ppm	0.50
Diazinon	< 0.14	ppm	0.50
Disulfoton	< 0.15	ppm	0.50
Methyl Parathion	< 0.14	ppm	0.50
Malathion	< 0.14	ppm	0.50
Parathion	< 0.12	ppm	0.50
Thiodan	< 0.02	ppm	0.50
Ethion	< 0.14	ppm	0.50
Trithion	< 0.15	ppm	0.50

P.O. Box 44220 • Madison, WI 53744-4220 • 800-483-5523 • www.harlan.com

harlan
Helping you do research better

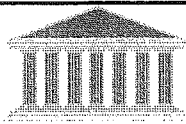
Harlan, Harlan Laboratories, Helping you do research better, and the Harlan logo are trademarks and trade names of Harlan Laboratories, Inc. © 2008 Harlan Laboratories, Inc.

SUITABLE FOR USE
REVIEWED BY: UG
DATE: 24 Aug 14

ANALYSIS REPORT

Revision 1

Page 1 of 1



KIRBY MEMORIAL HEALTH CENTER

71 N. Franklin Street, Wilkes-Barre, PA 18701 (570)823-5450 FAX(570)825-9926 PADEP#40-115

Calvert Laboratories Inc.
Scott Technology Park
100 Discovery Dr.
Olyphant, PA 18447
Attn: John Morahan

08/25/14

EXACT COPY
INITIAL: BK
DATE: 30 OCT 14

SAMPLE ID #: AA65087

SAMPLE DESCRIPTION: Room Viv 5

COLLECTION DATE: 08/12/14;09:00

SUBMITTAL DATE: 08/12/14;12:15

RESULTS:

ANALYSIS	UNITS	RESULT	QUANT LIMIT	ANALYSIS DATE	ANALYSIS TIME	ANALYST	METHOD	MCL*
Total Coliform	colifor m/ 100ml	Negative	1	08/13/14	15:04	KM	SM9223B	Neg
Coliform/Ecoli Incubation Start	---	---	---	08/12/14	15:04	KM	SM9223B	---
E. coli	E.coli/ 100 ml	Negative	1	08/13/14	15:04	KM	SM9223B	Neg
Nitrate	mg/l	0.72	0.20	08/12/14	19:54	AW	EPA 300.0	10.0
pH(at 15° C)	---	7.25**	---	08/12/14	13:36	KJ	SM4500H+B	3.5-8.5
Total Sodium	mg/l	20.0	3.0	08/19/14	12:48	JM	EPA 200.7	50.0

COMMENTS: *Reference: SOP#MNT-5

**Sample was received and analyzed outside of the 15 min. hold time. (KJ)

SUITABLE FOR USE
REVIEWED BY: [Signature]
DATE: 25 OCT 14

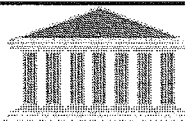
Reviewed, Approved, and Respectfully Submitted by:

Danielle Cappellini, B.Sc., M.H.A.
Director of Laboratory

ANALYSIS REPORT

Revision 1

Page 1 of 1



KIRBY MEMORIAL HEALTH CENTER

71 N. Franklin Street, Wilkes-Barre, PA 18701 (570)823-5450 FAX(570)825-9926 PADEP#40-115

Calvert Laboratories Inc.
Scott Technology Park
100 Discovery Dr.
Olyphant, PA 18447
Attn: John Morahan

08/25/14

EXACT COPY
INITIAL: AK
DATE: 30 Oct 14

SAMPLE ID #: AA65086

SAMPLE DESCRIPTION: Pharmacology 1 Sink

COLLECTION DATE: 08/12/14;09:00

SUBMITTAL DATE: 08/12/14;12:15

RESULTS:

MB 2 SCA 2014

ANALYSIS	UNITS	RESULT	QUANT LIMIT	ANALYSIS DATE	ANALYSIS TIME	ANALYST	METHOD	MCL*
Total Coliform	coliform/ 100ml	Negative	1	08/13/14	15:04	KM	SM9223B	Neg
Coliform/Ecoli Incubation Start	---	---	---	08/12/14	15:04	KM	SM9223B	---
E. coli	E.coli/ 100 ml	Negative	1	08/13/14	15:04	KM	SM9223B	Neg
Nitrate	mg/l	0.71	0.20	08/12/14	19:27	AW	EPA 300.0	10.0
pH(at 14°C)	---	7.22**	---	08/12/14	13:34	KJ	SM4500H+B	3.5-8.5
Total Sodium	mg/l	22.0	3.0	08/19/14	12:45	JM	EPA 200.7	50.0

COMMENTS: *Reference: SOP#MNT-5

**Sample was received and analyzed outside of the 15 min. hold time. (KJ)

SUITABLE FOR USE

REVIEWED BY: [Signature]
DATE: 2 Sept 14

Reviewed, Approved, and Respectfully Submitted by:

[Signature]
Danielle Cappellini, B.Sc., M.H.A.
Director of Laboratory

E. Appendix V – Protocol, Protocol Amendments and Deviations



Study Protocol

Title: Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)

Calvert Study No.: 1275RL35.001

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

LABORATORY

Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447 USA
Phone: 570 536 3411
Fax: 570 536 3450

www.calvertlabs.com

CORPORATE

1215 Crescent Green Dr
Suite 115
Cary, NC 27510
Phone: 919 854 4453
Fax: 919 854 2880

www.calvertholdings.com

I. Table of Contents

I. Table of Contents	2
II. Introduction	4
A. Title	4
B. Objective	4
C. Regulatory Compliance	4
D. Testing Guidelines	4
E. Declaration of Intent	4
F. Calvert Study Number	4
G. Testing Facility	5
H. Sponsor	5
I. Study Director	5
J. Study Monitor	5
K. Sponsor Representative	5
L. Principal Investigator for Dosing Formulation Analyses	6
M. Lead Quality Assurance - Calvert	6
N. Key Study Dates	6
III. Materials and Methods	7
A. Test Article and Vehicle	7
1. Test Article	7
2. Vehicle	7
3. Dose Preparation	7
4. Dose Formulation Sampling	8
5. Shipment and Analysis	8
6. Sample Batch Acceptance Criteria	10
7. Reserve Archive Samples	10
8. Accountability and Disposition	10
B. Test System (Animals and Animal Care)	10
1. Description	10
2. Rationale for Choice of Species and Number of Animals	11

3. Husbandry	11
4. Pre-study Health Screen and Selection Criteria	12
5. Assignment to Study Groups	13
6. Humane Care of Animals	13
C. Test Article Administration	13
1. Group Assignments and Dose Levels	13
2. Dosing	13
3. Justification for Route and Dose Levels	14
D. Method of Study Performance.....	14
E. Terminal Procedures.....	15
1. Termination.....	15
IV. Records and Reports.....	15
A. Data Collection and Statistical Analysis	15
B. Storage of Records	15
V. Miscellaneous	16
A. Confidentiality Statement	16
B. References.....	16
VI. Protocol Approval Signatures	17

II. Introduction

A. Title

Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)

B. Objective

To determine the potential effects of LT3114 administered intravenously on respiratory rate, tidal volume and minute volume in the conscious rat.

C. Regulatory Compliance

This study will be conducted in compliance with the Good Laboratory Practice Regulations 21 CFR Part 58, Japanese Ministry of Health, Labour and Welfare Ordinance 21 issued March 26, 1997, effective April 1, 1997, in accordance with OECD Principles of Good Laboratory Practice as revised on 26th November, 1997 [C(97)186/Final], and in accordance with the appropriate Standard Operating Procedures (SOP) of Calvert.

Study elements conducted by subcontractors will be performed in accordance with the appropriate Standard Operating Procedures of the subcontractor.

D. Testing Guidelines

FDA Guidance for Industry "S7A Safety Pharmacology Studies for Human Pharmaceuticals (ICH)" issued July 2001

E. Declaration of Intent

This study may be submitted to the United States Food and Drug Administration (FDA), the EMEA, an Organization for Economic Cooperation and Development (OECD) member country, and/or the Japanese Ministry of Health, Labor and Welfare (JMHLW).

F. Calvert Study Number

1275RL35.001

G. Testing Facility

Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

H. Sponsor

Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

I. Study Director

Brian Klatt, B.S.
Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447
Phone: (570) 585-2229
Fax: (570) 585-2383
E-mail: brian.klatt@calvertlabs.com

J. Study Monitor

Marlene W. Modi, Ph.D.
MWM Consulting Group, Inc. for Lpath Inc.
14 Joseph Court
Monmouth Junction, NJ 08852
Phone: (732) 274-9562
Mobile: (732) 822-6101
E-mail: mwmconsultinggroup@gmail.com

K. Sponsor Representative

Gary Woodnutt, PhD
Senior Vice President, Research
Lpath Inc.

4025 Sorrento Valley Blvd
San Diego, CA 92121
Phone: (858) 678-0800 ext 112
Email: rmatteo@lpath.com

L. Principal Investigator for Dosing Formulation Analyses

Victor T. Johnson, Ph.D.
Grants & Contracts PI/Study Director
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Fax: (866) 971-2128
Email: vjohnson@brt-labs.com

M. Lead Quality Assurance - Calvert

Darlene Gilpin, RQAP-GLP
Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447
Phone: (570) 585-2219
Fax: (570) 585-2383
E-mail: darlene.gilpin@calvertlabs.com

N. Key Study Dates

Proposed Study Initiation Date:	1 Aug 2014
Proposed First Day of Dosing:	15 Aug 2014
Proposed Experimental Completion Date:	22 Aug 2014

III. Materials and Methods

A. Test Article and Vehicle

1. Test Article

Identification:	LT3114
Lot/Batch No.:	103948
Physical Description:	Clear colorless liquid
Storage Conditions:	Stored refrigerated (2-8°C) and protected from light.
Characterization/ Stability:	Good Laboratory Practice (GLP) regulations require that all test articles be appropriately characterized for identity, purity, composition and stability. Compliance necessitates <u>documentation</u> that characterization and stability testing have been performed. The sponsor will provide the Study Director with a certificate of analysis/results of stability testing and whether analysis/testing was conducted under GLP/GMP. If characterization/stability is not provided or was not performed under GLP/GMP, this fact will be reflected in the compliance statement.

2. Vehicle

Identification:	LT3114 Diluent
Lot/Batch No.:	104011-001
Physical Description:	Clear colorless liquid
Storage Conditions:	2-8°C

3. Dose Preparation

The Sponsor will provide the test article already formulated at 32.1 mg/ml in sterile containers. Diluent will also be provided in separate sterile containers.

Diluent and test article need to come to room temperature before the formulation is prepared. Do not shake the formulation. Group 1 control animals will receive the diluent. When necessary depending on the dose level, doses will be prepared by dilution using the provided 32.1 mg/ml and diluent solutions under aseptic conditions. Before dose preparation, 2 mL of the 10% Polysorbate 80 solution provided, should be added aseptically to each liter of formulation buffer, also supplied, and mixed gently to ensure distribution throughout the volume of buffer. This diluent will then be used for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations.

4. Dose Formulation Sampling

On the first day of dosing, duplicate 1.0 mL samples of each test article formulation will be collected from the top, middle and bottom of each formulation to confirm the concentration and homogeneity. Duplicate samples (1.0 mL) from the middle of the vehicle will be collected to confirm the absence of test article. These samples will be stored at 2-8°C until shipment for dose formulation analysis.

These samples will be stored at 2-8°C until shipment for dose formulation analysis.

5. Shipment and Analysis

Dose formulation samples will be shipped to (on dry ice) and analyses will be performed under the direction of:

Victor T. Johnson, Ph.D.
Grants & Contracts PI/Study Director
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Fax: (866) 971-2128
Email: vjohnson@brt-labs.com

Good Laboratory Practice (GLP) regulations require that all formulated test articles be appropriately verified for concentration, homogeneity and stability.

Targeted Total Number of

Formulation Samples: 8 Collected, 4 (initial shipment)

Sample Storage:

Conditions: Refrigerated (2-8°C)

Sample Ship Date:

Upon approval of analytical laboratory

Testing Requirements:

Concentration and homogeneity

Analytical Methods:

To be added by protocol amendment

Automated Data

Collection Systems:

To be added by protocol amendment

If any samples are re-analyzed, the Study Director will be notified and details will be included in the analytical report.

All Quality Assurance Unit activities and Quality Assurance Unit reports for the bioanalytical analyses will be sent to the Calvert Study Director and Calvert Management. All protocol, SOP and GLP deviations associated with the bioanalytical analyses will be reported to the Calvert Study Director. The final bioanalytical report will have a GLP compliance statement and a Quality Assurance Unit statement which will have the dates of any inspections and the dates the QAU reports were sent to the Principal Investigator, Study Director, and Study Director's Management.

A report of the sample analysis procedures and results will be provided to the Study Director for inclusion as an appendix in the study report.

Remaining samples will be appropriately discarded prior to finalization of the study report.

6. *Sample Batch Acceptance Criteria*

For a sample batch run to be acceptable, a minimum of 75% of the QC samples for LT3114 at each concentration must be within $\pm 10.0\%$ from the nominal value. In addition, a minimum of 75% of the back-calculated CS values for LT-3114 must fall within 10.0% of their nominal values.

Concentration and homogeneity of dosing formulations will be determined by assay of formulation aliquots sampled from the top, middle and bottom locations of the dosing formulation container. Measured concentration of the dosing formulation will be acceptable if it is 85% to 115% of the nominal concentration. Formulated LT3114 will be considered homogeneous if the measured concentrations of the top, middle and bottom aliquots have a %CV value $\leq 10.0\%$.

In the event that a sample is outside of the acceptable specification range, the Study Director will justify the acceptability of the results or suggest re-analysis of the backup samples or retest the affected portion of the study.

7. *Reserve Archive Samples*

Since the study is of short duration, no retention sample of the test article will be maintained at Calvert.

8. *Accountability and Disposition*

Unused test article will be returned to the Sponsor designee at study completion (prior to issue of final report) or, if necessary, retained for use on related future studies. The Sponsor will be notified in advance of shipping and a transmittal letter will accompany the shipment. The material will be packed in a suitable container to maintain the conditions specified by the Sponsor during transit plus an adequate margin of safety to account for any possible transit delays.

B. Test System (Animals and Animal Care)

1. *Description*

Species:	Rat
Strain/Substrain:	CrI: CD [®] (SD) or according to supplier

Total Number:	Twenty four (Six per group)
Gender:	Male
Age Range:	7 to 10 weeks
Body Weight Range:	250-350 grams
Animal Source:	Charles River Laboratories or any other approved supplier and documented in the study records.
Experimental History:	Purpose-bred and experimentally naïve at the outset of the study.
Identification:	Ear tagged and color coded

2. *Rationale for Choice of Species and Number of Animals*

The rat is selected because the physiologic response of their airways is similar to that of a human (1, 2, 3). Previous research has been conducted with the test article in rats.

Based on Calvert's experience with this model and available literature on pulmonary effects of test drugs (4, 5, 6), the number of animals used in this study is the minimum number considered to provide valid scientific assessment.

3. *Husbandry*

Housing:	Animals will be group housed upon receipt in compliance with National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals will be kept will be documented in the study records. No other species will be kept in the same room. The experiment will be conducted in a room dedicated to this study and the dosing will occur in this room only.
Lighting:	12 hours light/12 hours dark.
Room Temperature:	20 to 26°C

Relative Humidity:	30-70%
Food:	All animals will have access to Certified Rodent Diet (TEKLAD) or equivalent <i>ad libitum</i> , unless otherwise specified. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants are known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which will be maintained in the Calvert archives and a copy will be included in the final report.
Water:	Water will be available <i>ad libitum</i> , to each animal via an automatic watering device, unless otherwise specified. The water is routinely analyzed for contaminants as per Calvert's SOPs. No contaminants are known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis will be maintained in the Calvert archives and a copy will be included in the final report.
Acclimation:	Study animals will be acclimated to their housing for a minimum of 5 days prior to their first day of dosing.

4. Pre-study Health Screen and Selection Criteria

All animals received for this study will be assessed as to their general health by a member of the veterinary staff or other suitably trained authorized personnel. During the acclimation period, each animal will be observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that are determined to be healthy and suitable for use will be assigned to this study. Any animals considered unacceptable for use in this study will be replaced with animals of similar age and weight from the same vendor.

5. Assignment to Study Groups

Animals will be selected by body weight and apparent good health and will be randomly assigned to study groups.

6. Humane Care of Animals

Treatment of animals will be in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, NRC, 2011, The National Academies Press). The Calvert Institutional Animal Care and Use Committee (IACUC) will approve the study protocol prior to finalization to insure compliance with acceptable standard animal welfare and humane care.

No alternative test systems exist which have been adequately validated to permit replacement of the use of live animals in this study. Every effort has been made to obtain the maximum amount of information while reducing to a minimum the number of animals required for this study. The assessment of pain and distress in study animals and the use or non-use of pain alleviating medications will be in accordance with Standard Operating Procedure VET-19, Criteria for Assessing Pain and Distress in Laboratory Animals. The study will be terminated in part or whole for humane reasons if unnecessary pain occurs. To the best of our knowledge, this study is not unnecessary or duplicative.

C. Test Article Administration

1. Group Assignments and Dose Levels

Group	Treatment	Dose (mg/kg)	Volume (mL/kg)	Concentration (mg/mL)	No. of Animals
					Male
1	Diluent	0	5	0	6
2	LT3114	10	5	2.0	6
3	LT3114	30	5	6	6
4	LT3114	50	5	10	6

2. Dosing

Route: Slow intravenous injection (~1 minute)

Frequency: Once

Procedure: Doses will be administered by intravenous injection via the lateral tail vein at a dose volume of 5 ml/kg. Each intravenous dose will be administered as a manual slow injection (~1 minute). Each animal will receive a mg/kg dose based on its most recent body weight.

3. *Justification for Route and Dose Levels*

The intravenous route was chosen as it is the intended route of administration in humans.

The dose levels were selected by the Sponsor based on in-life study results of a 14-day repeat intravenous dose range finding toxicity study of LT3114 in Sprague Dawley Rats. (Calvert study 0437RL35.002).

D. Method of Study Performance

Rats will be trained initially for two days in a head-out plethysmographic chamber for approximately 15 minutes each (± 5 minutes), before the day of the experiment. On the day of dosing, each animal will be weighed and placed in the plethysmographic chamber and allowed to stabilize for at least 5 minutes. Following stabilization, respiratory parameters (tidal volume, minute volume and respiratory rate) will be measured for 5 minutes continuously to obtain the predose baseline values. The rat will then be removed from the chamber and dosed as per group assignments. Following dosing, each animal will be returned to its designated plethysmograph chamber and the respiratory parameters will be measured 15 minutes (± 3 min), and 1, 2 and 4 hours (± 5 minutes) following administration of test article/vehicle. Animals will be allowed to stabilize in the plethysmograph for at least 5 minutes before each reading is taken. The five minute recording periods will begin at the designated time following dosing. The following parameters will be recorded using the DSI Dataquest Open ART (v 2.3) & PONEMAH Physiology Platform (v 4.2.0):

- Tidal Volume
- Minute Volume
- Respiratory rate

Any animal found dead or sacrificed prematurely during the study will be subjected to a necropsy, at the discretion of the Study Director. After opening the thoracic and

abdominal cavities, a macroscopic examination of the tissues will be performed. Any abnormalities will be recorded.

E. Terminal Procedures

1. Termination

a) Scheduled Sacrifice

All surviving animals will be euthanized by CO₂ asphyxiation following the 4 hour data collection. No necropsies will be performed, no tissues will be saved, and the carcasses will be discarded.

b) Unscheduled Sacrifice

Animals may be euthanized by CO₂ asphyxiation for humane reasons based upon criteria outlined in Calvert's SOP VET-14, "Criteria for Determination of Moribund Animals and for Sacrifice for Humane Reasons".

IV. Records and Reports

A. Data Collection and Statistical Analysis

Respiratory Data will be acquired and analyzed using DSI Dataquest Open ART (v 2.3) & PONEMAH Physiology Platform (v 4.2.0). Individual values of respiratory rate, tidal volume and minute volume for the test article-treated groups will be compared back to vehicle control and baseline using a two-way repeated measures ANOVA followed by a Bonferroni Multiple Comparison Test (SigmaStat, v2.03). Differences with p values ≤0.05 will be considered statistically significant. All other data will be hand recorded and manually entered into the report if applicable

B. Storage of Records

Test article preparation, test article tracking, in-life data, protocol, protocol amendments (if applicable), and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

All dose formulation data will be archived at Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560.

Six months following the submission of the audited draft report if there are no client comments generated by the Sponsor and/or Sponsor Representative, the Sponsor/Sponsor Representative will be notified and the report will be finalized and archived according to the terms stated in the protocol.

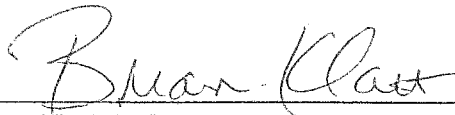
V. Miscellaneous

A. Confidentiality Statement

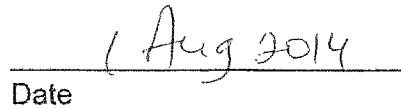
The information contained herein is for exclusive use of the intended recipient(s).

B. References

1. ICH S7A Safety Pharmacology Studies for Human Pharmaceuticals, July, 2001.
2. Peiser C, Undem BJ, Fischer A. Nociceptin effects in the airways. *Peptides.*, 2000 Jul;21(7):995-8.
3. Murphy, D.J. Assessment of respiratory function in safety pharmacology. *Fundam. Clin. Pharmacol.* 16: 183-196, 2002.
4. Schlenker, E.H., Shi, Y., Wipf, J., Martis, D.S., Kost, C.K. Fructose feeding and intermittent hypoxia affect ventilatory responsiveness to hypoxia and hypercapnia in rats. *J. Appl. Physiol.* 97: 1387-1394, 2004
5. Kaczynska, K. and Szereda-Przestaszewska, M. Clonidine-induced respiratory effects in anesthetized rats. *Exp. Physiol.* 91: 269-275 (2007).
6. Bailey, T.C., Maruscak, A.A., Petersen, A., White, S., Lewis, J.F., Veldhuizen, A.W. Physiological effects of oxidized exogenous surfactant in vivo: effects of high tidal volume and surfactant protein A. *Am. J. Physiol. Lung Cell. Mol. Physiol.* 291: L703-L709 (2006).

VI. Protocol Approval Signatures

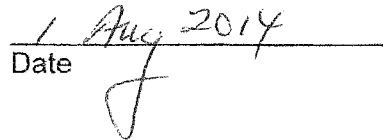
Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.



Date



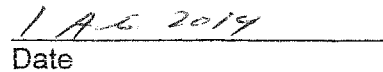
Steven D. Shreiner
Scientific Management
Calvert Laboratories, Inc.



Date



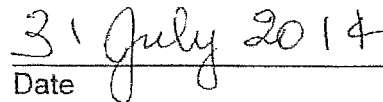
IACUC
Calvert Laboratories, Inc.



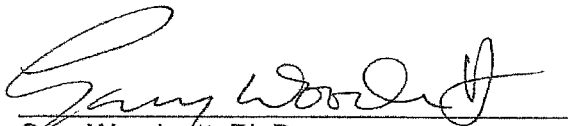
Date



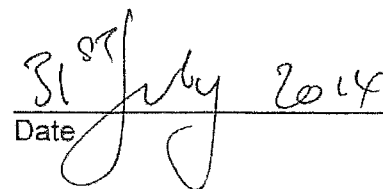
Marlene Modi, Ph.D.
Study Monitor



Date



Gary Woodnutt, Ph.D.
Sponsor Representative



Date

31 Jul 2014



PROTOCOL AMENDMENT

STUDY NUMBER: 1275RL35.001
STUDY TITLE: Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)
AMENDMENT NUMBER: 1
EFFECTIVE DATE: 14 Aug 2014

PROTOCOL SECTION CHANGE (Page 9 of 17):

III. Methods and Materials

A. Test Article and Vehicle

5. Shipment and Analysis

Targeted Total Number of
Formulation Samples: 8 **20** Collected, 4 **10** (initial shipment)

REASON FOR CHANGE:

The targeted number of samples was not changed to revise the homogeneity samples that will be collected.

EFFECTIVE DATE: 14 Aug 2014

PROTOCOL SECTION CHANGE (Page 4 of 17):

II. Introduction

C. Regulatory Compliance

This study will be conducted in compliance with the Good Laboratory Practice Regulations 21 CFR Part 58, ~~Japanese Ministry of Health, Labour and Welfare Ordinance 21 issued March 26, 1997, effective April 1, 1997,~~ **and** in accordance with OECD Principles of Good Laboratory Practice as revised on 26th November, 1997 [C(97)186/Final], and in accordance with the appropriate Standard Operating Procedures (SOP) of Calvert.

Study elements conducted by subcontractors will be performed in accordance with the appropriate Standard Operating Procedures of the subcontractor.

E. Declaration of Intent

This study may be submitted to the United States Food and Drug Administration (FDA), the EMEA, an Organization for Economic Cooperation and Development (OECD) member country, ~~and/or the Japanese Ministry of Health, Labor and Welfare (JMHLW).~~

REASON FOR CHANGE:

The reference to the Japanese regulations was removed to be consistent with the other studies being conducted at Calvert.

PROTOCOL SECTION CHANGE (Page 5 of 17):

II. Introduction

K. Sponsor Representative

Gary Woodnutt, PhD
Senior Vice President, Research
Lpath Inc.
4025 Sorrento Valley Blvd

San Diego, CA 92121

Phone: (858) 678-0800 ext 112

Email: ~~rmatteo@lpath.com~~ gwoodnutt@lpath.com

REASON FOR CHANGE:

To correct the contact information for the Sponsor Representative.

PROTOCOL SECTION CHANGE (Page 6 of 17):

II. Introduction

L. Principal Investigator for Dosing Formulation Analyses

~~Victor T. Johnson, Ph.D.~~ **Victor J. Johnson, Ph.D.**

PROTOCOL SECTION CHANGE (Page 8 of 17):

III. Methods and Materials

A. Test Article and Vehicle

5. Shipment and Analysis

Dose formulation samples will be shipped to (on dry ice) and analyses will be performed under the direction of:

~~Victor T. Johnson, Ph.D.~~ **Victor J. Johnson, Ph.D.**

REASON FOR CHANGE:

There was a typographical error in the middle initial of the Principal Investigator.

PROTOCOL SECTION CHANGE (Page 10 of 17):

III. Methods and Materials

A. Test Article and Vehicle

6. Sample Batch Acceptance Criteria (1st paragraph)

For a sample batch run to be acceptable, the following assay acceptance criteria will be met;

Calibration Curve Acceptance Criteria:

- Accuracy of standards: 20% of nominal (25% near LLOQ)
- Precision of the standards: %CV value $\leq 20.0\%$
- At least 75% of the standards (minimum of 6 points) are required to meet these criteria. Failed standards will be removed from the analysis.

QC Sample Acceptance Criteria:

- Accuracy of QC samples: 20% of nominal (25% near LLOQ)
- Precision of the QCs: %CV value $\leq 20.0\%$
- At least 67% of all QC samples are required to meet these criteria. In addition, at least 50% of the QC samples at each level are required to meet these criteria.

~~a minimum of 75% of the QC samples for LT3114 at each concentration must be within $\pm 10.0\%$ from the nominal value. In addition, a minimum of 75% of the back-calculated CS values for LT-3114 must fall within 10.0% of their nominal values.~~

REASON FOR CHANGE:

The sample batch criteria was changed to reflect the criteria used by the analytical lab.

SIGNATURES:



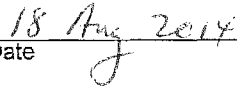
Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.



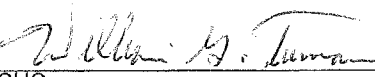
Date



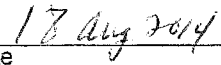
Steven D. Stenehjem
Scientific Management
Calvert Laboratories, Inc.



Date



William G. Luman
IACUC
Calvert Laboratories, Inc.



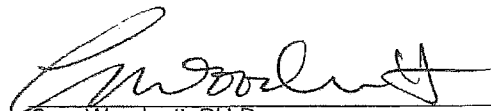
Date



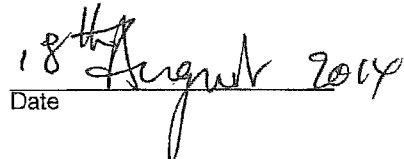
Marlene Modi, PH.D.
Study Monitor

16 August 2014

Date



Gary Woodnutt, PH.D.
Sponsor Representative



Date



PROTOCOL AMENDMENT

STUDY NUMBER: 1275RL35.001
STUDY TITLE: Pulmonary Assessment of LT3114 in the Conscious Rat (GLP)
AMENDMENT NUMBER: 2
EFFECTIVE DATE: 31 Oct 2014

PROTOCOL SECTION CHANGE (Page 9 of 17):

III. Methods and Materials

A. Test Article and Vehicle

5. Shipment and Analysis

Analytical Methods:	To be added by protocol amendment BRT 20140629-A
Automated Data Collection Systems:	To be added by protocol amendment Spectramax 340 microplate spectrophotometer and/or Spectramax M2e microplate spectrophotometer Analysis of the ELISA absorbance data is performed with Softmax Pro v2.2.1 and/or v5.0.1

**Analysis of dose formulation
acceptance criteria is performed using
Microsoft Excel v2013**

**Dose formulation stability has been conducted under BRT
20140629-A**

REASON FOR CHANGE:

To add the Analytical Method and Automated Data Collection Systems and to include a reference to dose formulation stability.

PROTOCOL SECTION CHANGE (Page 10 of 17):

III. Methods and Materials

A. Test Article and Vehicle

6. Sample Batch Acceptance Criteria (2nd paragraph)

Concentration and homogeneity of dosing formulations will be determined by assay of formulation aliquots sampled from the top, middle and bottom locations of the dosing formulation container. Measured concentration of the dosing formulation will be acceptable if it is ~~85%~~ **80%** to ~~115%~~ **120%** of the nominal concentration. Formulated LT3114 will be considered homogeneous if the measured concentrations of the top, middle and bottom aliquots have a %CV value ~~≤10.0%~~ **20%**.

REASON FOR CHANGE:

To update the dose formulation acceptance criteria

SIGNATURES:



Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.

31 Oct 2014
Date



Steven D. Shukla
Scientific Management
Calvert Laboratories, Inc.

31 Oct 2014
Date



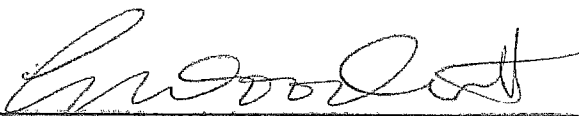
William G. Turner
IACUC
Calvert Laboratories, Inc.

31 Oct 2014
Date



Marlene Modi, PH.D.
Study Monitor

31 October 2014
Date



Gary Woodnutt, PH.D.
Sponsor Representative

31st October 2014
Date

Protocol Deviations

1. The relative humidity in the animal room deviated from the protocol-stated range of 30 to 70% on one occasion to a high of 76%. The room temperature in the animal room deviated from the protocol-stated range of 20 to 26°C on one occasion to a low of 19°C. The deviations were not of a magnitude or duration that would have affected the outcome of the study.
2. The protocol stated that the dose formulation samples were to be shipped to the analytical laboratory on dry ice. The samples were stored refrigerated and were shipped on wet ice packs. Since the samples were sent under the correct storage condition, this deviation had no impact on the outcome of the study.